Learner's Workbook

Worksheets and Exercises

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Worksheet 1.1: Introduction to Space Mission Design

Name: ___________________________ Date: ___________

Part A: The 17 SE Processes -> SpaceCDF Mapping

For each NASA SE process, identify which SpaceCDF feature (tab, button, or panel) implements it. Mark "Manual" if it requires human judgment.

#	SE Process	SpaceCDF Feature	Manual?
1	Stakeholder Expectations Definition		
2	Technical Requirements Definition		
3	Logical Decomposition		
4	Design Solution Definition		
5	Product Implementation		
6	Product Integration		
7	Product Verification		
8	Product Validation		
9	Product Transition		
10	Technical Planning		
11	Requirements Management		
12	Interface Management		
13	Technical Risk Management		
14	Configuration Management		
15	Technical Data Management		
16	Technical Assessment		
17	Decision Analysis		

Discussion question: Which processes are well-supported by the tool? Which ones require the team's judgment and cannot be automated?

Part B: Lifecycle Phase Identification

For each activity below, identify which NASA lifecycle phase it belongs to and what review gate follows:

Activity	NASA Phase	Review Gate
"We need to monitor sea ice thickness"		
Writing system-level requirements		
Selecting a reaction wheel vendor		
Running thermal-vacuum tests		
Operating the satellite in orbit		
Decommissioning the satellite		
Preliminary orbit trade study		
Detailed structural analysis (FEA)		

Part C: CDF Position Interaction Map

Draw lines between positions that must interact to resolve each design issue:

Issue 1: The payload needs 50W but the solar array can only provide 30W.
- Positions involved: ____

Issue 2: The reaction wheel vibration exceeds the payload stability requirement.
- Positions involved: ____

Issue 3: The selected orbit requires a propulsive deorbit but there's no mass budget for propulsion.
- Positions involved: ____

Issue 4: The X-band downlink interferes with the payload's sensitive receiver.
- Positions involved: ____

Part D: SpaceCDF Exploration

Navigate through the SpaceCDF tool and answer:

1. How many KPI cards are shown on the Dashboard? List them:

____

2. What are the MCR exit criteria in the Gate Review tab?

____

3. What positions are available when starting a session?

____

4. What mission types can be selected in the Requirements step?

____

Notes & Discussion Points

Use this space to record key points from group discussion:

____

____

____

____

Worksheet 1.2: Mission Need & Stakeholder Analysis

Name: ___________________________ Date: ___________

Part A: Stakeholder Matrix

Identify at least 4 stakeholders for your team's mission. For each, capture their needs, constraints, and priority.

#	Stakeholder	Role	Key Needs	Constraints They Impose	Priority (P/S)
1					
2					
3					
4					
5					

Discussion: Which stakeholder has the most influence? Which has needs that conflict with others?

Part B: Objective Quality Checklist

Write 2-3 mission objectives below. Then score each against the checklist.

Objective 1: ____

____

Criterion	Yes/No	Notes
States WHAT, not HOW?		
Has a measurable criterion with number + unit?		
Is achievable with current technology?		
Is relevant to the problem statement?		
Is traceable to a stakeholder need?		
Is classified as primary or secondary?		

Objective 2: ____

____

Criterion	Yes/No	Notes
States WHAT, not HOW?		
Has a measurable criterion with number + unit?		
Is achievable with current technology?		
Is relevant to the problem statement?		
Is traceable to a stakeholder need?		
Is classified as primary or secondary?		

Part C: Problem Statement Writing

Write your team's mission problem statement below. It should answer:
- What is the problem (capability gap)?
- Who is affected?
- What is the impact of not solving it?
- What constraints exist?

Problem Statement:

____

____

____

____

Peer review: Exchange with another team. Does their problem statement prescribe a solution (HOW)? Mark any solution language.

Part D: MoE -> MoP -> TPM Chain

For one of your objectives, trace the measurement hierarchy:

Objective: ____

MoE (operational effectiveness from user perspective):

____

MoP (technical performance measure):

____

TPM (design parameter to track during development):

____

Part E: SpaceCDF Tool Entry

In SpaceCDF Step 1 (Mission Need), enter your problem statement, stakeholders, and objectives. Then answer:

1. Did the tool auto-detect your mission type from the objectives? What type did it suggest?

____

2. How many stakeholders did you enter? Are there any the tool doesn't let you capture?

____

3. What measurable criteria did you enter for each objective?

____

Notes

____

____

____

Worksheet 2.1: Requirements Engineering

Name: ___________________________ Date: ___________

Part A: Requirement Quality Assessment

Evaluate each requirement against the SMART checklist. Mark Pass (P) or Fail (F). Rewrite any failures.

#	Requirement	S	M	A	R	T	HOW?	Rewrite
1	"The spacecraft shall operate at 500 km altitude"							
2	"The system shall achieve GSD <= 10 m at nadir"							
3	"The EPS shall use triple-junction GaAs cells"							
4	"The system shall have a design lifetime of at least 3 years"							
5	"The structure shall be made of aluminium 7075"							

Part B: Requirement Hierarchy

For your mission, write one requirement at each level:

Mission Requirement: ____

System Requirement (derived): ____

Subsystem Requirement (derived): ____

Draw the traceability: which objective does the mission requirement trace to? Which function does the subsystem requirement derive from?

Part C: Splitting Compound Requirements

Split this compound requirement into individually testable statements:

"The system shall provide 10m GSD multispectral imagery with 5-day revisit, 24-hour data latency, and positive power margin in all operational modes including eclipse."

#	Split Requirement	Verification Method (A/T/R/I)
1		
2		
3		
4		
5		

Part D: SpaceCDF Requirements Generation

1. In SpaceCDF, click "Generate from Objectives." How many requirements were generated? ____

2. Use the level filter -- how many at each level?
   - Mission: ____ System: ____ Subsystem: ____

3. Find one requirement that says HOW not WHAT. Write it and your correction:
   - Original: ____
   - Corrected: ____

4. Accept at least 5 requirements and reject at least 1. Record your decisions:

Req ID	Action	Rationale

Worksheet 2.2: Functional Decomposition

Name: ___________________________ Date: ___________

Part A: Function Tree

Draw your mission's function tree (minimum 3 levels). Use the format:

```
F-001: [Primary mission function]
  ??? F-002: [Subfunction 1]
  ??? F-003: [Subfunction 2]
  ??? F-004: [Subfunction 3]
F-005: [Universal function 1]
F-006: [Universal function 2]
...
```

Your function tree:

Part B: Function-to-Requirement Traceability

For each leaf function, write one derived requirement:

Function ID	Function Name	Allocated To	Derived Requirement

Part C: Multi-Allocation Analysis

Identify at least one function that could be allocated to more than one subsystem:

Function: ____

Option A allocation: ____

Option B allocation: ____

Interface created: ____

Recommended boundary: ____

Part D: Coverage Check

After entering functions in SpaceCDF, check for coverage gaps:

- How many leaf functions have NO linked requirements? ____
- List them: ____
- What requirements would you derive for these?

Worksheet 2.3: Interface Management

Name: ___________________________ Date: ___________

Part A: Interface Identification

List the 5 most critical interfaces in your design:

#	Subsystem A	Subsystem B	Interface Types	Key Concern
1				
2				
3				
4				
5				

Part B: Interface Requirements

Write 2 formal interface requirements for your most critical interface pair:

Interface: ________ <-> ________

IR-001: ____

IR-002: ____

Are these verifiable? By what method (A/T/R/I)? _______________

Part C: Conflict Resolution

From the SpaceCDF Interface Matrix, identify one conflict (red border):

Conflict: ____

Severity: Critical / Major / Minor

Resolution options considered:
1. ____
2. ____
3. ____

Selected resolution: ____

Rationale: ____

Part D: Discussion

Which interface do you expect will cause the most problems during integration? Why?

____

____

Worksheet 2.4: Design Budgets

Name: ____________________________ Date: ___________

Part A: Mass Budget

Complete from your SpaceCDF design:

Subsystem	CBE Mass (kg)	% of Dry	Equipment Margin (20%)	MEV (kg)
Payload				
EPS				
AOCS				
TTC				
OBC				
Thermal				
Structure				
Harness				
Propulsion				
Dry Total		100%		
System margin (20%)				
Dry MEV				
Propellant				
Wet Mass				
Allocation				
Margin				

Part B: Power Budget by Mode

Subsystem	Safe (W)	Idle (W)	Imaging (W)	Downlink (W)	Eclipse (W)
OBC					
AOCS					
Payload					
TTC					
Thermal					
Total					
Duty cycle %					

Orbit-average power: P_avg = _____ W

SA power needed: P_SA = P_peak + recharge = _____ + _____ = _____ W

SA BOL (with degradation): _____ W

Part C: Budget Health Check

From the SpaceCDF Dashboard, record:

Budget	Margin	Status (G/A/R)
Mass	____%	
Power	____%	
Link	____ dB	
Cost	____ MEUR vs ____ ceiling	
Pointing	____° vs ____° req	

Which budget is tightest? ____

What design change would improve it? ____

Worksheet 3.1: Orbit Design & Selection

Name: ____________________________ Date: ___________

Part A: Orbital Mechanics Calculations

Your selected orbit: Altitude h = ____ km, Inclination i = ____°, Type: ________

Compute (show working):

1. Semi-major axis: a = R_E + h = 6371 + ____ = ____ km = ____ m

2. Orbital period: T = 2pi?(a^3/mu) = _____ s = _____ min

3. Orbital velocity: v = ?(mu/a) = _____ m/s

4. Eclipse fraction: f = (1/pi)×arccos(?(1-(R_E/a)^2)) = _____

5. Sunlight time per orbit: t_sun = T × (1-f) = _____ min

6. Eclipse time per orbit: t_ecl = T × f = _____ min

Part B: Orbit Trade Matrix

From the SpaceCDF orbit trade advisor, record the top 3 candidates:

Rank	Orbit	Alt (km)	Inc (°)	GSD (m)	Revisit (d)	Lifetime (yr)	5yr?	Score
1								
2								
3								

Selected orbit: ____ Rationale: ____

Part C: Debris Compliance

For your selected orbit:
- Natural lifetime: ____ years
- FCC 5-year compliant? Y / N
- IADC 25-year compliant? Y / N
- Propulsion needed for deorbit? Y / N
- If yes, estimated deorbit ?V: ____ m/s

Part D: Discussion

What would happen to your mission if you moved 100 km higher? 100 km lower?

____

____

Worksheet 3.2: Payload & Communications

Name: ___________________________ Date: ___________

Part A: Payload Sizing (Optical missions -- skip if non-optical)

GSD requirement: ____ m Altitude: ____ km Wavelength: 0.55 mum

1. Diffraction-limited GSD: GSD_diff = 1.22 × lambda × h / D = _____ m (for D = ____ m)

2. Required aperture for target GSD: D >= 1.22 × lambda × h / GSD = _____ m

3. Payload mass estimate: M = 20 × D^1.5 + 2 = _____ kg

4. Payload power estimate: P = 3 × M = _____ W

Part B: Link Budget

Complete the link budget for your mission's primary downlink:

#	Parameter	Value	Unit
1	TX Power		dBW
2	TX Antenna Gain		dBi
3	TX Losses		dB
4	EIRP		dBW
5	Free Space Path Loss		dB
6	Atmospheric Loss		dB
7	Pointing Loss		dB
8	RX Antenna Gain		dBi
9	System G/T		dB/K
10	C/N0		dBHz
11	Data Rate (10log10)		dBbps
12	Eb/N0 available		dB
13	Eb/N0 required		dB
14	Implementation margin		dB
15	LINK MARGIN		dB

Does the link close (>= 3 dB)? Y / N

Part C: Frequency Band Selection

License type chosen: Amateur / Experimental / Commercial

Band selected: _____ Frequency: _____ MHz

Rationale: ____

Filing required: ____

Part D: SpaceCDF Link Budget Tool

Open the Link Budget tab and enter your parameters. Compare to your hand calculation:

Parameter	Hand Calc	SpaceCDF	Difference
EIRP			
FSPL			
Link Margin			

Worksheet 3.3: Power, AOCS, & Thermal

Name: ___________________________ Date: ___________

Part A: Solar Array Sizing (show calculations)

1. Peak sunlight power demand: P_peak = _____ W (from mode: _________)
2. Eclipse power demand: P_eclipse = _____ W
3. Eclipse time: t_ecl = _____ min; Sunlight time: t_sun = _____ min
4. Recharge power: P_rech = (P_ecl × t_ecl)/(t_sun × 0.9) = _____ W
5. SA EOL required: P_SA = P_peak + P_rech = _____ W
6. EOL degradation factor (3yr, 2.5%/yr): (1-0.025)^3 = _____
7. SA BOL required: P_SA_BOL = P_SA_EOL / factor = _____ W
8. SA area: A = P_BOL / (0.295 × 1361 × 0.85) = _____ m^2
9. SA type needed: Body-mounted / Single deploy / Dual deploy

Part B: Battery Sizing

1. Eclipse energy: E = P_ecl × t_ecl/60 = _____ Wh
2. Battery capacity: C = E / (DoD × eta) = _____ / (0.3 × 0.95) = _____ Wh
3. Battery mass: m = C / 150 = _____ kg

Part C: AOCS Selection

Pointing requirement: _____ °

AOCS architecture selected: (tick one)
- [ ] Passive magnetic (>5°)
- [ ] Magnetorquers only (2-5°)
- [ ] Reaction wheels + MTQ (0.1-2°)
- [ ] Fine pointing: RW + star tracker + MTQ (<0.1°)

Part D: Pointing Error Budget

Error Source	Value (°)	Value^2
Sensor accuracy		
Actuator resolution		
Alignment knowledge		
Thermal distortion		
Jitter		
Orbit knowledge		
RSS Total	=?(?) =	

Requirement: _____ ° Margin: _____ ° (_____ %)

Part E: Thermal Check

From SpaceCDF Dashboard, record:
- Predicted hot case temp: _____ °C
- Predicted cold case temp: _____ °C
- Component max operating: _____ °C -> Hot margin: _____ °C (need >=5°C)
- Component min operating: _____ °C -> Cold margin: _____ °C (need >=5°C)

Worksheet 3.4: Structure, Propulsion, & Equipment Selection

Name: ___________________________ Date: ___________

Part A: CDS Compliance Check

CDS Requirement	Your Design	Compliant?
Form factor (___U) dimensions		
Max mass (___kg)		
Rail material (anodised Al)		
Deployment switches (min 2)		
RBF pin provided		
No protrusions beyond rails		
CG within 2 cm of geometric centre		

Part B: Propulsion Decision

Does your mission need propulsion?
- Orbit altitude: _____ km
- Natural lifetime: _____ years
- FCC 5-yr compliant without propulsion? Y / N
- Orbit maintenance needed? Y / N
- Constellation phasing needed? Y / N

Decision: Propulsion / No propulsion

If propulsion: Type: _________ Isp: _____ s ?V needed: _____ m/s
Propellant mass: m = m_dry × (e^(?V/(Isp×9.81)) - 1) = _____ kg

Part C: Equipment Selection Log

Category	Component Selected	Manufacturer	Mass (kg)	Power (W)	Cost (kEUR)	Qty
EPS Board						
Battery						
Solar Panels						
OBC						
Reaction Wheel						×
Magnetorquer						×
Star Tracker						
Sun Sensor						×
Transponder						
Antenna						
Structure						
TOTAL						

Parametric estimate: _____ kg Selected total: _____ kg Difference: _____ kg

Part D: Component Trade Study

Subsystem traded: _________ Components compared: 3

Criterion	Weight	Option A: _______	Option B: _______	Option C: _______
Mass	0.20			
Power	0.15			
Cost	0.20			
TRL	0.15			
Heritage	0.15			
Performance	0.15			

| Weighted Score | | | | |

Winner: _________ Rationale: ____

Worksheet 4.1: Equipment Selection & Integration

Name: ___________________________ Date: ___________

Part A: Compatibility Verification

For each selected component, verify interface compatibility:

Component	Bus Voltage OK?	Protocol Compatible?	RF Band Match?	Notes

Part B: Budget Tracking (from Equipment Browser)

After all selections, record the live budget bar values:

- Total selected mass: _____ kg
- Total selected power: _____ W
- Total selected cost: _____ kEUR
- Mass allocation: _____ kg -> Margin: _____ kg (_____ %)
- Volume utilisation: _____ % of ___U form factor

Does everything fit? Y / N -- If not, what would you change?

____

Part C: Incompatibility Found and Resolved

Describe one incompatibility detected during selection and how you resolved it:

Issue: ____

Components involved: ____

Resolution: ____

Worksheet 4.2: Verification & Validation Planning

Name: ___________________________ Date: ___________

Part A: V&V Method Assignment

For 10 key requirements, assign method (A/T/R/I), phase, and level:

#	Requirement ID	Requirement Text (abbreviated)	Method	Phase	Level	Justification
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						

Part B: Dual-Method Requirements

Identify 3 requirements that need BOTH analysis AND test:

Requirement	Analysis (Phase B)	Test (Phase C/D)

Part C: Environmental Test Sequence

For your selected launch vehicle, specify the test sequence:

1. [ ] Initial functional test
2. [ ] ______________ vibration (levels: _____ gRMS, axes: 3, duration: ___/axis)
3. [ ] Post-vibe functional test
4. [ ] Thermal vacuum (hot: +_____°C, cold: -_____°C, cycles: ____)
5. [ ] Post-TVAC functional test
6. [ ] EMC test (required? Y/N)
7. [ ] Mass measurement
8. [ ] Deployer fit check

Launch vehicle: _______________ PUG used: _______________

Worksheet 4.3: Risk Management

Name: ___________________________ Date: ___________

Part A: Risk Register

#	Risk Description	L (1-5)	C (1-5)	Score	Category	Mitigation Strategy	Owner
1							
2							
3							
4							
5							

Scoring: 1-4 Low (accept), 5-9 Medium (mitigate if practical), 10-15 High (must mitigate), 16-25 Critical (redesign)

Part B: Single-Point Failure Analysis

Component	Failure Mode	Effect on Mission	SPF?	Acceptable?	Mitigation

Part C: Discussion

What is the highest-risk item in your design? What would it take to bring the risk to an acceptable level?

____

____

____

Worksheet 4.4: Cost Estimation

Name: ___________________________ Date: ___________

Part A: Mission Cost Estimate

WBS Element	Parametric Est. (kEUR)	Bottom-Up Est. (kEUR)	Notes
Programme Management			
Systems Engineering			
Mission Assurance			
Payload			
Bus Hardware			
Integration & Test			
Software (FSW + GSW)			
Launch			
Ground Segment			
Operations (__ years)			
TOTAL			

P50 estimate: _____ kEUR P80 estimate (×1.3): _____ kEUR

Part B: Constellation Cost (if applicable)

Number of satellites: ____ Learning rate: _____%

Unit #	Unit Cost (kEUR)	Cumulative (kEUR)
1		
2		
4		
8		
Total (N=___)		

+ Spares (15%): ____ units × _____ kEUR = _____ kEUR
+ Launch: ____ sats × _____ kEUR = _____ kEUR
+ Ground + Ops: _____ kEUR

Total constellation cost: _____ kEUR = _____ MEUR

Part C: Cost Drivers

Top 3 cost drivers in your mission:
1. ____
2. ____
3. ____

What would you change to reduce cost by 20%? ____

Worksheet 5.1: Gate Review Preparation

Name: ___________________________ Date: ___________

Part A: MCR Exit Criteria Status

From SpaceCDF Gate Review tab, record each criterion:

#	Criterion	Status (Pass/Fail/Manual)	Evidence	If Fail: Fix Action
EC-01	Mission need defined			
EC-02	Stakeholders identified			
EC-03	Objectives with criteria			
EC-04	Alternatives considered			
EC-05	Concept justified			
EC-06	ConOps documented			
EC-07	Feasible concept			
EC-08	Sustainable (debris)			
EC-09	Requirements traceable			
EC-10	Interfaces managed			

Overall readiness: READY / NOT READY Must-pass met: ___/___

Part B: MCR Presentation Outline

Prepare your team's 12-minute MCR presentation:

1. Mission Need (2 min): ____
2. Why Space? (2 min): ____
3. System Design (3 min): ____
4. Feasibility (2 min): ____
5. Risks (2 min): ____
6. Actions (1 min): ____

Part C: Action Items from Review

#	Action	Responsible	Deadline	Acceptance Criteria
1				
2				
3				

Board decision: GO / NO GO / GO with actions

Worksheet 5.2: Regulatory & Licensing

Name: ____________________________ Date: ___________

Part A: Filing Requirements Checklist

Tick all that apply to your mission:

- [ ] ISED spectrum licence (CPC-2-6-02) -- all missions transmitting
- [ ] ITU API/coordination/notification -- commercial/experimental
- [ ] IARU amateur coordination -- if using amateur bands
- [ ] RSSSA operating licence -- if ANY Earth imaging capability
- [ ] Export permit -- if controlled components or US launch
- [ ] Controlled Goods registration -- if handling US-origin controlled items
- [ ] COPUOS registration -- all missions (post-launch)
- [ ] FCC Part 25 -- if US-licensed or US launch
- [ ] End-of-life compliance report -- all missions

Part B: Licensing Decision

Your license type: Amateur / Experimental / Commercial

Frequency band: _________ Data rate: _____ Mbps

ITU filing needed? Y / N Estimated timeline: _____ months

RSSSA needed? Y / N Reason: ____

Export control concerns? Y / N Components of concern: _______________

Part C: Regulatory Timeline

Working backward from your target launch date:

Milestone	Date (L minus)	Filing	Authority
L - __ months		ITU API	ISED->ITU
L - __ months		ISED spectrum	ISED
L - __ months		RSSSA (if needed)	Global Affairs
L - __ months		Export permits	Global Affairs
L - __ months		IARU coord (if amateur)	RAC->IARU
L + 1 month		COPUOS registration	Global Affairs

Critical path filing: ____

Part D: SpaceCDF Templates

Generate these from the Exports tab and note TBD fields:

Document	# Auto-filled fields	# TBD fields	Key TBDs
ITU API			
RSSSA			
Export Assessment			
COPUOS Registration			
End-of-Life Report			

Worksheet 5.3: Launch Integration

Name: ___________________________ Date: ___________

Part A: Launch Provider Selection

From SpaceCDF Launch Selector, record top 3 options:

Provider	Vehicle	Price	Capacity	Your Mass	Fits?	Deployer

Selected: _________ Rationale: ____

Mass allocation from provider: _____ kg -> Mass margin: _____ kg (_____ %)

Part B: ICD Compliance Checklist

ICD Requirement	Your Design	Compliant?	Evidence
Dimensions within deployer envelope			
Mass within deployer limit			
CG within 2 cm of geometric centre			
2+ deployment switches on rails			
RBF pin provided			
No protrusions beyond rails (stowed)			
Battery <= 50% SoC at delivery			
No RF emissions in deployer			
Antennas stowed clear of deployer			

Part C: Environmental Test Specification

From launch vehicle PUG, specify your test levels:

Random vibration: _____ gRMS, _____ min/axis, 3 axes
Sine vibration: _____ g, _____ - _____ Hz
Shock: _____ g at _____ Hz
TVAC hot: predicted max + 10°C = _____ °C
TVAC cold: predicted min - 10°C = _____ °C
TVAC cycles: ____

Part D: Discussion

What ITAR implications arise from launching your Canadian CubeSat on a US vehicle?

____

____

Worksheet 5.4: Optimisation & Final Design

Name: ____________________________ Date: ___________

Part A: Sensitivity Analysis

From SpaceCDF Optimizer (or Trade Studies), identify which parameters most influence your design:

Variable	mu* (importance)	sigma (non-linearity)	Classification
			Key driver / Negligible / Investigate

Most important parameter: ____

Action: ____

Part B: Final Design Summary

Parameter	Value	Unit	Margin	Status
Dry mass		kg	___%	G/A/R
Wet mass		kg		
SA power EOL		W	___%	G/A/R
Battery capacity		Wh		
Link margin		dB		G/A/R
Pointing accuracy		°	___%	G/A/R
Total cost		MEUR		
Data balance		GB/day surplus		G/A/R
Debris compliance		/100		G/A/R

Part C: Documents Generated

Tick each document generated from SpaceCDF:

- [ ] MRD (Mission Requirements Document)
- [ ] Technical Specification
- [ ] Verification Plan
- [ ] ConOps Document
- [ ] SEMP
- [ ] Risk Management Plan
- [ ] IRD
- [ ] Test Plan
- [ ] BOM
- [ ] ITU Filing Template
- [ ] RSSSA Filing
- [ ] End-of-Life Report
- [ ] COPUOS Registration

Part D: Lessons Learned

1. What was the most challenging design decision?

____

2. What would you do differently if starting over?

____

3. What does the design need next (Phase B activities)?

4. What was the most valuable feature of the tool?

5. What feature was missing that you needed?

Final Presentation Peer Review (completed by reviewing team)

Criterion	Score (/10)	Comments
Problem clearly defined		
Design feasibility (budgets close)		
Trade study rigour		
Requirements quality		
Completeness		
Communication clarity		
Risk awareness		
TOTAL	/70	